Ambient Systems

Machine Learning Technology for Smart Buildings

Managing Indoor Environments

Building owners and operators often struggle to managing indoor environments in real-time because of three underlying challenges.

1. **Air Quality**: although building operators follow establish guidelines, many are operating with a black box when it comes to understanding current quality of indoor air.

2. **Complexity**: building operations consist of complex legacy systems of machinery that controls cooling, ventilation, purification, and other processes.

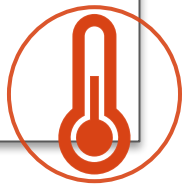
3. **Many uses**: various building uses will each create a slightly different demand in building's systems.

Air Quality

There are three essential parameters that need to be measured and managed in order to create an optimal indoor environment based on unique building operations and occupant needs.

- Impacts comfort and productivity
- Correlated with humidity
- Energy intensive

Temperature



- Must be within a specific healthy range
- Correlated with temperature
- Impacts spread of contaminants

Relative Humidity



- Organic: pollen, mold, bacteria
- Inorganic: VOCs, SVOCs, Ozone (O_3)
- CO_2
- Other airborne pathogens

Contaminants



Ambient Systems

Ambient is a one stop shop for building owners and operators who want to manage indoor environment and air quality seamlessly and effectively.

Building Components

- Mechanical
- Electrical
- Sensors & Controls
- Ambient Sensors*
- Ambient Controls*

Ambient Platform**

Understand
Identify and measure parameters important for building operations

Optimize
Fine-tune building systems to achieve desired goals

Improve
Identify opportunities for equipment upgrades

Applications

Healthcare
Minimize airborne infections

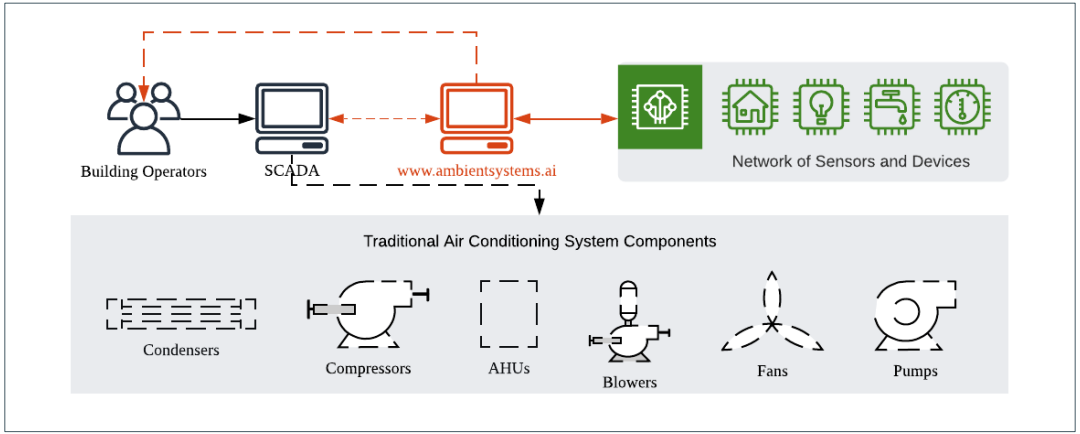
Offices
Increase worker comfort and productivity

Museums
Preserve invaluable artifacts

Vertical Farms
Increase crop yield

* Based on unique building needs. General input parameters: temperature, relative humidity, CO_2, contaminants, and user feedback.
** Platform outputs are available online, delivered by email/message notifications, or connected to your bdenuilding's SCADA system

Ambient Systems
High-level Schematic



What makes Ambient different?

- Technology tailored for unique building operations and occupant needs
- Seamlessly connects to the building's existing SCADA system and provides notifications directly to the operators
- Adds an additional layer of intelligence using AI-enabled optimization models
- Managed on the cloud and available across your devices
- Enabled by a network of interconnected sensors and devices (Internet-of-Things)
- Leverages NUS intellectual property to minimize maintenance and energy costs of sensors and devices

Thank You

Contact